

Production Guide for Print - Common

The Production Guide for Print - Common is a guide for users supporting Print in a production environment. This document provides an overview of the print process and information on what, how, and when to monitor. In addition, this document provides information on troubleshooting print errors.

This is part of a three document set. Refer to companion documents—

- Production Guide for Print - Exceed
- Production Guide for Print - POINT IN

Revision date - February 19, 2008



Table of Contents

Chapter 1: Automated Cycles.....	3
Exceed and POINT IN Document Processing.....	3
GenTrn Process	4
GenData Process.....	4
GenData Output Files.....	5
GenPrint Process.....	6
GenWIP Process.....	7
GenWIP Incomplete Transactions	7
GenArc Archive Process.....	8
Printing the Document	8
Chapter 2: Common Document Production Monitoring	9
Documaker RP Error Monitoring	9
Technical Support	10
Documaker RP WIP Purge Program	10
Chapter 3: Recovery.....	12
Automated Cycle Recovery	12
Identifying transactions	12
Chapter 4: Document Print Error Handling	14
Errors Produced by Document Print Web Service (.NET Version)	14
Errors Produced by Document Print Web Service (Non .NET Version).....	15
Errors Produced by Documaker RP Programs	15
Tracing SOAP Messages Sent to and/or Received by the Web service (.NET Version)	16
Tracing SOAP Messages Sent to and/or Received by the Web Service (non- .NET Version)	17
Controlling the Deletion of the Temporary Directories Created by Document Print (.NET Version).....	19
Updating the DOCPRINT.INI File	20
Appendix A: Print Monitoring Checklist	21
Exceed PEI	21
Appendix B: References.....	22
Reference Material Applicable to This Manual.....	22

Chapter 1: Automated Cycles

Exceed and POINT IN Document Processing

The error and log file figure below shows the Documaker RP product and relationships between processes, error files, and log files for generating printable documents.

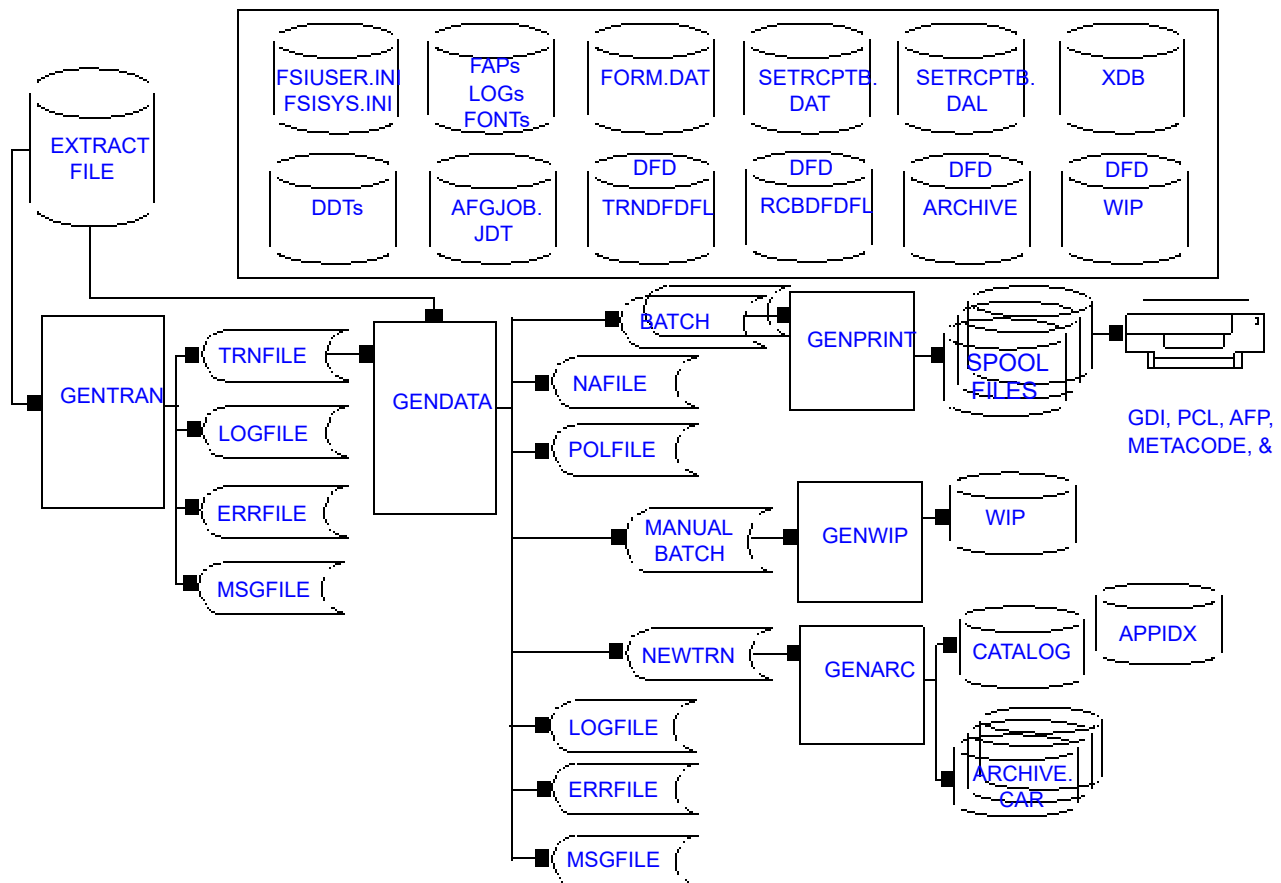


Figure 1. Document Production Program Flow and File Relationships

Document Production Process Executable Name	
GENTRN	GENTNW32
GENDATA	GENDAW32
GENPRINT	GENPTW32
GENWIP	GENWPW32
GENARC	GENACW32

GenTrn Process

GenTrn takes the source data, which is stored in extract files, and creates a list of the transactions stored in a transaction file (TRNFILE). This transaction list is then used by the GenData program as it processes transactions.

GenTrn uses settings in the FSISYS.INI and TRNDFDFL.DFD files to determine how to process the transactions. These files provide the GenTrn program with information about the format and structure of the extract file, such as how to determine where each new record starts.

ERRORFILES GenTrn produces an error file to note any transactions it could not process correctly. This file allows you to find and correct processing errors. The information is recorded by transaction. This file can be viewed using a text editor.

LOGFILES When GenTrn is run, the program creates log files that record each transaction processed. This file can be viewed using a text editor.

Note: A log file and error file is created with each GenTrn process. Previous files are overwritten each time. It is important that a backup be made before running GenTrn.

GenData Process

The GenData program takes the TRNFILE, created by the GenTrn program, and applies processing rules to those transactions and data. The GenData program creates batch files, NAFILE.DAT and POLFILE.DAT, for the GenPrint program. It also creates a manual batch file, MANUAL.BCH, for the GenWIP program and ERRORBATCH that contains all transactions that had errors posted. The output from GenData is also used by GenArc to archive forms and data.

GenData Output Files

NAFILE.DAT GenData creates a NAFILE.DAT file, commonly referred to as the NA file, to store image and variable field information. The GenPrint program uses this file along with the POLFILE.DAT file, produced by GenData, to print the forms.

If the data is incomplete and GenData cannot complete the form, it creates a manual batch file. The GenWIP program then creates separate DAT and POL files for each incomplete transaction. These files provide the entry system with the information it needs to open the form so a data entry operator can add the missing data. This is a semicolon-delimited text file.

POLFILE.DAT POLFILE.DAT, commonly referred to as the POL file, defines the form set used for a specific transaction. GenData creates this file, which is used by GenPrint, GenWIP, and GenArc. For instance, if the data is complete, GenData creates an NA file and a POL file. These files are used by GenPrint, along with the batch files, to produce the print-ready file.

If the data is incomplete and GenData cannot prepare the form for printing, it creates a manual batch file. The GenWIP program then creates separate files to house the NA file and POL file data for each transaction. These files provide the entry system with information needed to open the form, allowing missing data to be added by data entry operators.

NEWTRN.DAT GenData creates the NEWTRN.DAT file, which contains all transactions that processed without error. This file tells GenArc where to find data in the NAFILE.DAT file and which forms to use in the POLFILE.DAT file.

BATCHFILES GenData creates files with the extension BCH, called batch files. It lists the transactions to be included in each batch as specified in the FSISYS.INI file settings. Batch files are used as trigger files by GenPrint and GenWIP. Batch files indicate which transactions should be printed in a given batch job. GenPrint uses batch files to print completed forms. The GenData program also creates manual batch files, which record incomplete transactions. These manual batch files are used by GenWIP. Transactions that had errors are placed in the ERROR.BCH file.

MANUAL.BCH GenData creates the MANUAL.BCH file if it is unable to complete the processing of a form set. Typically this occurs because information

is missing. This file is then used by GenWIP, allowing data entry operators to manually complete and resubmit the form for processing.

ERRORBATCH GenData creates error batch files when it detects an error. In contrast to manual batch files, these errors cannot be corrected using GenWIP. These errors should be reported to the Document Production system administrator for resolution.

UPDATED LOG ERROR AND MESSAGE FILES As GenData processes information, it updates the log, error, and message files. You can review these files using a text editor.

GenPrint Process

GenData produces batch files, NAFILE.DAT, and POLFILE.DAT, which the GenPrint program uses to create printed forms.

UPDATED LOG AND ERROR FILES As the GenPrint program processes information, it updates log and error files. You can review these files using a text editor.

OMR marks are used to mark the beginning and end of formsets, sorted by LOB, MCO, and ZIP code. Barcodes can also indicate when to insert pre-printed materials before or after specified pages. A print device (such as Bell & Howell Mailstar) scans each page for the marks and performs specific actions based on the marks. Document Production has code to add OMR marks to forms as images after the formsets are built. This code is flexible enough to allow off

customization by scripting in support files such as FSISYS.INI, SET-RCPTB.DAT, and others.

The OMR routine is a callback function inside of GenPrint. It reads the INI files to determine when and where to put the OMR marks. It also checks for mailers, the first image triggered, indicating when one formset is complete and the next formset begins.

Inserts are controlled by the OMRHopper section of the INI file. This applies to all batches. Inserts cannot be turned off based on the batch. Inserts are controlled based on the recipient code. This is handled in the SORT_CREBRE section, using RECIP =. All recipients not listed would be turned off.

GenWIP Process

GenWIP receives information about incomplete transactions from the GenData program. This information is stored in manual batch (MANUAL.BCH) files.

Data can be corrected and missing information supplied to complete the form in Documaker Workstation. Once completed, the form can be resubmitted for processing by the GenData program.

GenWIP Incomplete Transactions

GenWIP creates the following files as it prepares incomplete transactions for further processing with the entry system:

- | | |
|---------------------|---|
| WIP.DBF | WIP.DBF contains information about the incomplete transactions extracted by GenWIP from the NAFILE.DAT file and the POLFILE.DAT file created by GenData. The WIP.MDX file serves as an index to this file. |
| WIP.MDX | This is an index to the WIP.DBF file. |
| 00000001.DAT | Uses the MANUAL.BCH file produced by GenData. GenWIP creates a separate data file for each incomplete transaction from the NAFILE.DAT file. These files are numbered and have an extension of .DAT. They are like NAFILE.DAT except there is only one transaction per file. |
| 00000001.POL | Uses the MANUAL.BCH file produced by GenData. The GenWIP program creates a separate POL file for each incomplete transaction from the POLFILE.DAT file. These files are numbered to correspond with their matching data file and contain information about the form set on which the system should place the data. They are like POLFILE.DAT except there is only one transaction per file. |

Note: *The naming convention of the DAT and POL files are controlled by the WIP.DFD files.*

- | | |
|-------------------|--|
| UNIQUE.DBF | The UNIQUE.DBF file contains the last number for the WIP file that was created. When a WIP file is created, a unique number is generated so that no WIP file is overwritten. This file, therefore, should not be modified, renamed or deleted. The highest number that will be generated for WIP files is FFFFFFFF, which is 4,294,967,295. After this number, the counter resets to 00000001. |
|-------------------|--|

Note: The Skywire default configuration for the WIP database is DBF and MDX files. Your environment may be configured for a different database.

GenWIP uses this information to create separate data and form information files from the incomplete transaction information it receives from the GenData program.

GenArc Archive Process

GenArc creates and uses these files as it archives information:

APPIDX.DBF The APPIDX.DBF file is created and updated by GenArc. This is the main index for the archive.

APPIDX.DFD This is an external file that defines the archive index record that is contained in APPIDX.DBF.

ARCHIVE.CAR GenArc creates CAR files in which archived forms and data are stored. ARCHIVE.CAR is an example of a generated CAR file. It is possible to have multiple CAR files depending on your INI file configuration.

APPIDX.MDX Index to the APPIDX.DBF file.

Note: The Skywire default configuration for the ARCHIVE database is DBF and MDX files. Your environment may be configured for a different database.

Printing the Document

GenPrint takes information produced by GenData and creates a printer spool file for use with PCL, AFP, Metacode, and Post Script printers. Print spool files may be transmitted to a third party vendor for printing, adding inserts, and mailing.

Chapter 2: Common Document Production Monitoring

The following section describes Documaker RP files that require routine review to capture errors, transaction failures, or other system discrepancies that could occur during normal processing.

Documaker RP Error Monitoring

ERROR FILES

GenTrn produces an error file to note any transaction that could not be processed correctly. Because this file is overwritten with each batch cycle, it must be monitored every day. The information is recorded by transaction. This file can be viewed using a text editor.

The Error Batch is created by GenData if an error is detected. Unlike manual batch files, these errors cannot be corrected using the GenWIP program.

These errors should be reported to the Document Production system administrator. Typically, these transactions are stripped from the TRNFILE and saved for later processing. The system administrator or a designated person will research the issue and correct the problem. Error file messages usually diagnose problems encountered during the processing run such as:

- Invalid data recognition
- Improper options
- Input/output errors
- Resource validation

The problem can be corrected by updating any number of Document Production resource files, including, but not limited to, a DDT file,

table data, or an INI setting. If the issue is corrected within the Document Production system, the administrator can reprocess the policies that previously had errors by running a separate batch process.

Sometimes the problem lies within bad or invalid data in the extract file and must be corrected on the host first. When corrected, the policy transaction is submitted in a subsequent batch.

LOGFILES When GenTrn is run, the program creates log files that record each transaction processed. This file can be viewed using a text editor.

Log file messages are informational and generally not tied to the success or failure of a job. They provide information about each transaction as it proceeds through the processing cycle such as:

- Transaction IDs processed
- Start
- End
- Elapsed time of the run
- Transaction counts
- Statistics

Message numbers are associated with the Translation file (TRANSLAT.INI). This file contains all the static text that is merged with the variable information to produce the message written into log and error files.

GENWIP GenWIP should be monitored each day to find documents that have missing data because these documents will not be processed until corrected.

Technical Support

Documaker RP WIP Purge Program

Whenever Document Print transactions are sent to WIP for batch printing, they are stored in the WIP database with a status code of "B." During the next batch cycle run, Document Production will read the WIP database for transactions with this status code, merge them with the batch transactions, and change the status code to "SR,"

indicating they have been serviced. A batch purge program called WSPPGW32 is run at the end of the cycle to delete the “SR” status code entries from the WIP database. This prevents the batch cycle from processing these transactions again in the next batch run. To implement this process, include WSPPGW32.exe at the end of your batch cycle.

Chapter 3: Recovery

Automated Cycle Recovery

If the Automated Cycle fails during processing, recovery begins with troubleshooting applicable output files (See [Automated Cycles](#) on page 3):

Error File	ERRFILE.DAT
Log File	LOGFILE.DAT
Message File	MSGFILE
Transaction File	TRNFILE.DAT

Transactions that failed can be identified and isolated (removed from Transaction File) for future processing. The TRNFILE.DAT created during Gentran lists all transactions available for processing. The LOGFILE.DAT contains activity for each transaction. If a transaction halts during processing, the last item (in order of processing) can be found here. The LOGFILE.DAT has print request IDs and could contain policy numbers to aid in identification.

Following removal of the applicable transaction(s), processing should continue by restarting Gendata. Transactions can be removed by editing the TRNFILE.DAT (See [Automated Cycles](#) on page 3).

Identifying transactions

Transactions are identified by print request IDs. Once the new version of the transaction files is saved, processing can continue by restarting the Gendata Executable (Gendaw32). The transaction removed must be documented and forwarded to the implementation team for handling.

If the Automated Cycle fails and the source of the problem and/or transaction is not easily identified in the resource files mentioned

above, further technical review may be necessary (refer to the ***Doc-***
umaker RP Technical User Guide).

If the process fails during Genprint, the same resource files will be used to determine the root cause.

At this point, removing a transaction may not be an option. An error during Genprint usually results from invalid INI settings that control printer ports, batch selections, fonts, etc. Recovery will require technical intervention to resolve Genprint issues. Valid settings can be found in the Documaker RP Technical User Guide, specifically, items that reference FSISYS.INI updates, batches, printers and so forth.

Chapter 4: Document Print Error Handling

Document Print produces two types of errors, errors produced by the Web service and errors produced by Documaker RP programs that are called by the Web service.

Errors Produced by Document Print Web Service (.NET Version)

When the Document Print Web service encounters an error, the error and error message, along with a return code of -1, are thrown back to the calling application for the calling application to catch and display in the browser. Additional information about the exception may be found in the prtreqid.err file that is created in the root of the temporary directory. Below is sample of the content for this error file:

METHOD: RetrieveFile

MESSAGE: Error in RetrieveFile(). Print File is empty. File name = E:\DocProdiUT\MSTRRES\DOCPRINT\POINTDMS\f29575\Print\

SOURCE: DocPrintService.BusinessLayer

STACK TRACE: at CSC.DocPrintService.BusinessLogic.FileHdlr.RetrieveFile(String strPath, String strFileExt)at CSC.DocPrintService.BusinessLogic.CreateDocument.CreateDocument(String strReqType, String strUniqueId, String strEnvrnmnt, String strXmlChgData, Byte[] bytExtrData, String& strRetMimeType)

Where:

- **Method** - Method Name the exception occurred in.
- **Message** - Error message produced by the exception
- **Source** - Source module the exception occurred in
- **Stack Trace** - execution path when the exception occurred

Errors Produced by Document Print Web Service (Non .NET Version)

In release 2004.3 and prior the errors produced by the Document Print Web service are logged to the event log under application and a return code of -1 is returned back to the calling application. To view the error in the event log follow the steps below:

- 1 Select **Start > Settings > Control Panel > Administrative Tools > Event Viewer**.
- 2 Select **Application**.

Errors Produced by Documaker RP Programs

The Document Print Web service can also generate an error when one of the GEN programs returns an error. Error messages produced by the GEN programs can be found in the errfile.dat file located in the DATA folder under the temporary folder created for that transaction. Additional information about the error may be found in the msgfile.dat file in the same location.

Tracing SOAP Messages Sent to and/or Received by the Web service (.NET Version)

At times there may be a need to verify the information in the SOAP message that is sent to the Document Print Web service or to see if the calling application is actually making contact with the Web service. This can be accomplished by activating SOAP trace. The steps below show how to turn on SOAP trace:

- 1 Open the web.config file in \DocPrintService\WebServer\WebService
- 2 Set enable = "true"

```
<microsoft.web.services2>
  <diagnostics>
    <trace enabled="false"
      input="InputTrace.webinfo"
      output="OutputTrace.webinfo" />
  </diagnostics>
</microsoft.web.services2>
```

The SOAP request message is logged in the input file name and the SOAP response message is logged in the output file name. Below is an example of the SOAP trace output:

InputTrace.webinfo file:

```
<soap:Body>
  <callDocumentProduction xmlns="http://www.csc.com/DocPrintService">
    <strReqType>HTML</strReqType>
    <strUniqueId>1659894112</strUniqueId>
    <strEnvrnmnt>EXCEED</strEnvrnmnt> <strRetMIMEType />
  </callDocumentProduction>
</soap:Body>
```

OutputTrace.webinfo file:

```
<soap:Body>
  <callDocumentProductionResponse xmlns="http://www.csc.com/DocPrintService">
    <callDocumentProductionResult>0</callDocumentProductionResult>
    <strRetMIMEType>HTM</strRetMIMEType> </callDocumentProductionResponse>
  </soap:Body>
```

Note: The files created by the SOAP trace will contain an accumulation of all SOAP messages.

Tracing SOAP Messages Sent to and/or Received by the Web Service (non- .NET Version)

In release 2004.3 and prior, tracing the SOAP message requires the use of the SOAP Toolkit 3.0 Trace Utility. To start the trace utility follow the steps below:

- 1 Select **Start > Programs > Microsoft SOAP Toolkit Version 3.0 > Trace Utility**.
- 2 Select either **File > New > Formatted Trace** or enter **Ctrl+F**
- 3 Select **OK** The following screen will appear.

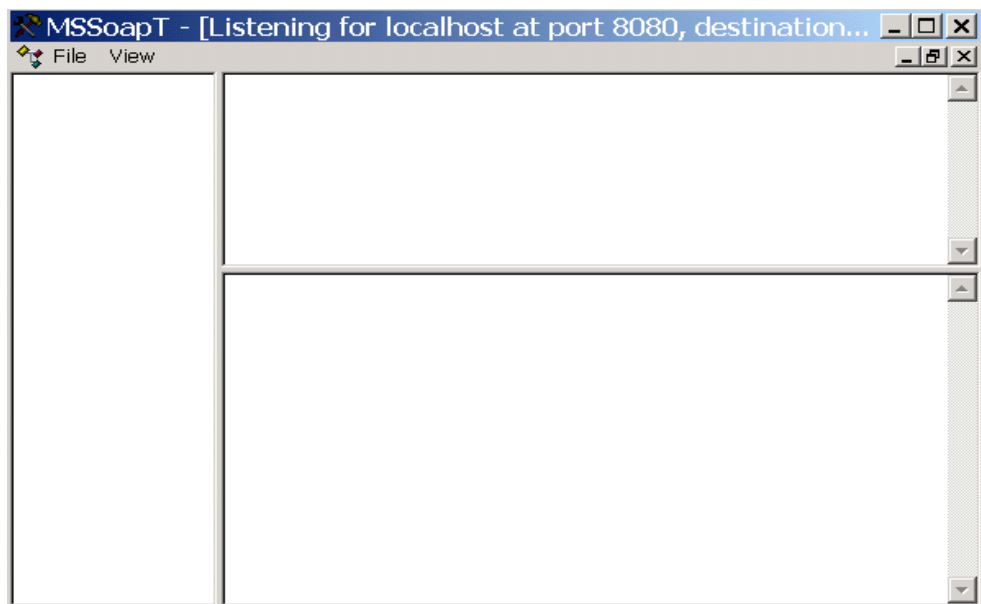


Figure 2. Formatted Trace

After starting the trace utility, you will need to add port 8080 to the DOCPRINT.WSDL file so the SOAP message will go through the trace utility. Follow the steps below:

- 1 Open the DOCPRINT.WSDL in Notepad. The file can be found in the following path: <drive>:\<target>\Mstrres\DocPrint\Resource.

- 2 At the bottom of the file under <service name='DocPrint' >, update the URL path by adding “:8080” after the IP address for the Web service you want to trace.

When Document Print runs, the trace utility will populate the left panel with the IP address for each request. Expand the IP address node, and click on the Message node. This will display the SOAP message in the two right panes. The request message is shown in the top right panel and the response message is shown in the bottom right panel.

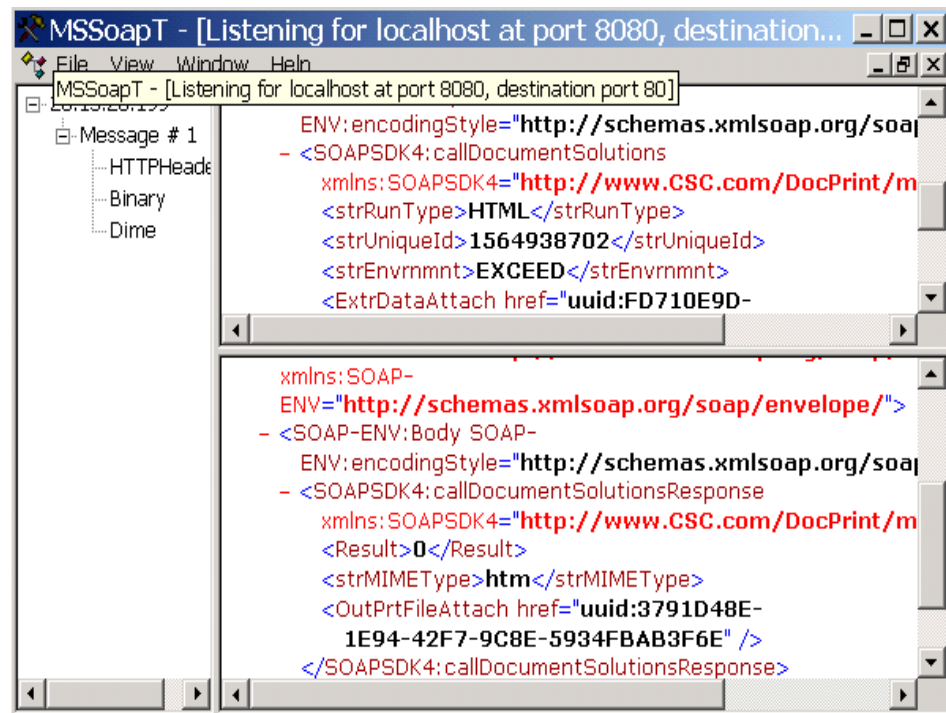


Figure 3. Trace Utility

Note: After you have closed the trace utility, you must remove the port value from the WSDL file or Document Print will not work.

Controlling the Deletion of the Temporary Directories Created by Document Print (.NET Version)

Whenever the first call is made to Document Print for a transaction, Document Print will create a temporary directory so the batch programs (GEN executables) can process. After the transaction has been sent to Archive or WIP, the temporary directory is automatically deleted. If an error occurred in the process, the temporary directory is not deleted for research purposes. However there may be a need to not delete the temporary directory when no errors occurred. The reason can be for researching a problem with a document containing incorrect data.

No matter what the reason is for the deletion of the temporary directory, it can be controlled by a setting in the DOCPRINT.INI file. To turn on or off this process, follow the steps below:

- 1 Open DOCPRINT.INI, located in <drive>:\<target>\DocPrintService\AppServer\Bin.
- 2 Under the appropriate environment section, set the following option:

DELETE_TEMP = Y or N

Where:

Y = Delete temporary directory on successful completion
(default)

N = Do not delete temporary directory on successful completion

- 3 You will need to restart IIS for the change to take.

Note: You can use the DPCLEANUTIL utility job to delete temporary directories based on a time elapsed from creation date. Refer to the Document Print Technical User Guide for more information.

Updating the DOCPRINT.INI File

Whenever a modification is made to the DOCPRINT.INI file, you must restart IIS. IIS caches this file, and any changes made to this file will not take until IIS is restarted.

To restart IIS, open a command prompt and enter “iisreset.”

Appendix A: Print Monitoring Checklist

Exceed PEI

- Exceed PEI maintaining processing queues
- Monitoring error queues
- Recovery procedures
- Back up and restore procedures

Appendix B: References

Reference Material Applicable to This Manual

- 1 *Production Guide for Print - Exceed*
- 2 *Production Guide for Print - POINT IN*
- 3 *Document Production Technical User Guide*
- 4 “Scheduled Processing” in *Print Enabling Interface (PEI) Operations Guide*
- 5 *Business Framework Operations Guide* for tips on managing CICS transactions and backup procedures for servers and databases
- 6 *Document Production Document Print Technical User Guide*
- 7 *POINT IN Media Center*